ORIGINAL PAPER

Trends and Patterns of Late and Unstaged Lung, Colorectal, Female Breast, and Prostate Cancers among American Indians in the Northern Plains, 2002–2009

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Abstract: **Introduction.** We investigated incidence and staging patterns of prostate, female breast, lung, and colorectal cancer among American Indians/Alaska Natives (AI/ANs) and non-Hispanic Whites (NHWs) in the Northern Plains. **Methods.** Cancer registry data (2002–2009) from Nebraska, North Dakota, and South Dakota were analyzed. Incidence rates were calculated and multivariate logistic regression analyses identified factors associated with unstaged versus staged and late-stage cancer cases *versus* early. **Results.** The incidence rate was higher among AI/ANs than NHWs for lung cancer (92.2 vs. 60.6 per 100,000). Compared with NHWs, AI/ANs were 2.0 times more likely to receive an unstaged diagnosis and 1.2 times more likely to receive a late-stage diagnosis. AI/ANs were significantly more likely than NHWs to receive an unstaged diagnosis. **Discussion.** Increased efforts are needed to reduce unstaged and late-stage diagnoses among Northern Plains AIs. Efforts to promote early detection of cancer should target younger AI/ANs.

Key words: Cancer, late stage, access, American Indians.

Within the American Indian/Alaska Native (AI/AN) population in the United States, Northern Plains AI/AN men and women, respectively, have the highest and second highest cancer incidence rates.[1] Moreover, Northern Plains AI/ANs have the highest cancer mortality rate among the AI/AN population nationally (275.7 per 100,000 population), a higher rate than the national AI/AN rate (165.6 per 100,000 population) and for all races combined in the United States (200.9 per 100,000 population).[2] There is a dearth of research on staging trends and patterns in this population.

In 2005, the Great Plains Tribal Chairmen's Health Board (GPTCHB) established the Northern Plains Comprehensive Control Program to address cancer disparities among

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AI/AN communities in the four-state Northern Plains region (North Dakota, South Dakota, Iowa, and Nebraska), which is regularly served by the GPTCHB. There are 17 federally recognized tribes in the region. According to the U.S. Census 2010, there were a total of 179,396 individuals who identified themselves as AI/AN in this four-state region.[3] In this four-state region, AI/ANs have considerably lower socioeconomic status and less access to health care compared to Non-Hispanic whites (NHWs) in the region and AI/ANs in other U.S. regions combined. For example, only 42% of Northern Plains AI/ANs have above a high school education compared to 59% of Northern Plains NHW counterparts and 48% of AI/ANs in other U.S regions combined. About 32% of Northern Plains AI/AN adults have no health insurance compared to 10% of Northern Plains NHW counterparts and 23% of AI/ANs in other U.S. regions combined. It is important to note that not all AI/ANs are eligible to receive care through Indian Health Service (IHS) and/or tribal health care facilities. For instance, individuals that identify as AI/ANs, but are not enrolled members may not be eligible to receive care through these facilities.

In 2007, the Northern Plains Tribal Cancer Data Initiative was developed to improve access and use of cancer data. The tribal oversight committee identified cancer stage patterns and trends as some of the primary issues to be investigated under this initiative. Cancer stage at diagnosis is the primary factor in cancer prognosis and affects treatment options. Patients diagnosed at a later stage have fewer treatment options and less favorable outcomes.[4] Past research has shown that cancer patients who are members of ethnic or racial minorities, uninsured, receiving Medicaid, of lower socioeconomic status (SES), living far from hospitals or primary care providers, and living in physician shortage areas are significantly more likely than their counterparts to be diagnosed with late-stage cancer.[5–10] Despite the importance of staging, a substantial proportion of those who are diagnosed with cancer are not staged.[11] Reasons for unstaged cancer cases are not well established.

During the last two decades there have been considerable changes in clinical and public health practices that may have influenced the likelihood that a cancer patient receives staging.[12–14] Among those changes was an increase in cancer screening use, advancements in diagnostic techniques, and availability of new treatment options. The continuing effort of cancer registry programs to improve the thoroughness and accuracy of data may also have contributed to lowering the number of unstaged cases found in the registry data over time. For example, one of the data standard goals established by the North American Association of Central Cancer Registries is reducing the proportion of cases ascertained from death certificates only. Although death certificates are important to identify additional cases that were not reported by medical facilities, they provide little or no information about the stage or treatment of cancer. If a case is identified only from a death certificate, then stage information must be recorded as "unknown" in the cancer registry database.

To our knowledge, most published studies that have included analysis of unstaged cancer cases were limited to one cancer site and/or used only data prior to 2000. In this study, we analyzed the data from three cancer registries that cover the Northern Plains region to understand better the patterns and trends of unstaged and late-stage cancers among AI/ANs compared with non-Hispanic Whites (NHWs). We focused on the four most common types of cancer, which collectively account for over half of cancers expected to be diagnosed in the United States in 2015 and which are ranked the top four cancers in the AI/AN population.[15]

Methods

We used cancer registry data (2002–2009) from Nebraska, North Dakota, and South Dakota to identify incident prostate, female breast, lung, and colorectal cancer cases diagnosed in AI/ANs and NHWs age 19 years and older from 2002 to 2009. Iowa data were not used because the number of cancer cases reported among AI/AN residents was too small for state-level analysis. For example, during the 1998–2007 period, the Iowa Cancer Registry recorded 221 total AI/AN cases (personal communication Michelle West, 2010). At the time of the data request submitted to state cancer registries, 2009 data were the most recently available data. The cutoff of 2002 was used because the data quality prior to 2002 was less optimal for one of the state cancer registries. Because prostate, female breast, lung, and colorectal cancers primarily affect adults, we only used data on individuals 19 years or older. Cases were identified using the *International Classification of Diseases for Oncology, Third Edition* codes of C619 for prostate cancer; C500–C506, C508, and C509 for female breast cancer; C340–C343, C348, and C349 for lung cancer; and C180–C189, C199, C209, and C260 for colorectal cancer. Within the same cancer site (e.g., breast), if there was more than one tumor per patient, the first diagnosed tumor was included in the analysis. We used the first diagnosis because we were interested in analyzing patient-level data, not tumor-level data.

The main dependent (outcome) variable is the status of staging (staged *vs*. unstaged). The SEER summary stage variables were used to categorize tumors into staged (*in situ*, local, and regional and distant stages) and unstaged (unknown stage). When describing early versus late staging in this study, we considered *in situ* and local as "early" and regional and distant as "late" stages. We used reporting source information to identify cases that were ascertained based on death certificate information only. Independent variables include the time period at diagnosis, sociodemographic characteristics (age, gender, race/ethnicity, socioeconomic status [SES]), and residence rurality. The time period at diagnosis was divided into early (2002–2005) and recent (2006–2009) periods. Age at the time of cancer diagnosis was categorized in 10-year increments, <55, 55–64, 65–74, 75–84, and 85 years or older. We used the following race/ethnicity categories: NHW and AI/AN (including both Hispanic and non-Hispanic ethnicity).

Information about SES is not consistently collected by the state cancer registries, so the U.S. Census Bureau block group information was used to create an SES index, using validated methods.[4,16,17] This index was created by using seven indicator variables that were combined into individual block group SES values using principal component analysis. The SES index component scores for the census block groups were then sorted and categorized into quintiles to assign SES value (1 to 5, with 1 indicating the lowest SES level). The seven indicators used were proportion of the population that is working class, proportion of the population older than 16 years in the workforce without a job, proportion of the population below 200% of the poverty level, an education index (based on proportion of the population with a college, high school, or less than high school education), median household income, median rent, and median house value. The average of the surrounding blocks was used to create a value for block groups where the median household income, median rent, or median house value were zero.

Patients were identified as living in urban, small urban, or rural counties by the county of residence at the time of diagnosis. A dichotomous definition of rural and urban communities[6,7,9,18] may mask important differences in cancer incidence and mortality rates and misclassify a vast proportion of the population that may be affected by proximity to an urban area.[8] Therefore, Sankaranarayanan *et al.*,[10,19] in an analysis of Nebraska colorectal cancer data, proposed a classification system that incorporates urban influence based on the U.S. Department of Agriculture's 12-level Urban Influence Codes (UIC).[16,20,21] Specifically, the UIC classification accounts for a county's adjacency to metropolitan areas, which may reflect the accessibility and availability of local health services.[22] In our previous studies using Nebraska cancer registry data, we first classified cases into the following four residence-county groups using a modified UIC code: (1) urban, including either large metropolitan counties (population of one million or more) or small metropolitan counties (population 50,000 to less than a million); (2) small-urban, including micropolitan or urban cluster counties either adjacent or not adjacent to a small or large metropolitan area (population 10,000 to less than 50,000); (3) rural, including non-metropolitan counties not adjacent to a metropolitan or micropolitan area and containing a town of 2,500–9,999 population; and (4) remote rural, including non-metropolitan counties not adjacent to a metropolitan or micropolitan area and containing a town of less than 2,500 population (personal communication with Timothy Parker, Sociologist, Economic Research Service, U.S. Department of Agriculture. USDA, July 8, 2008).

To examine the trend of unstaged cancer cases, unstaged incidence rates were calculated per 100,000 population and age-adjusted by the direct method using the 2000 U.S. standard population.[23,24] Incidence rates, rate ratios, and 95% confidence intervals (CIs) were calculated. To identify factors predicting unstaged versus staged cancer cases, multivariate logistic regression analyses were conducted. Odds ratios (ORs) and 95% CIs were obtained to evaluate the associations between patient demographic characteristics, time period, and staging outcome. We used SAS (SAS Institute Inc., Cary, NC, U.S.A.) software Version 9.2, and consider the level of significance to be at $p<.05$.

Results

Table 1 shows the characteristics of AI/AN and NHW individuals diagnosed with cancer. Compared with NHW cancer patients, AI/AN cancer patients were significantly younger and more likely to live in rural and lower SES communities. On average AI/ANs were 8.5 years younger than NHWs when they received a colorectal cancer diagnosis. About half of AI/AN cancer patients lived in communities classified as having a very low SES, compared with 2% or less of NHWs. More than half (52.1%) of AI/AN lung cancer patients were females, compared with 43.2% of NHWs.

Table 2 shows variations in age-adjusted incidence rates and rate ratios (RRs) across the three states. Among the NHW population, the incidence rates were similar across the states for each of the four cancers examined. However, AI/AN incidence rates varied

Table 1.

CHARACTERISTICS OF AI/ANS AND NHWS INDIVIDUALS WITH LUNG, COLORECTAL, FEMALE BREAST, AND PROSTATE CANCER: NE, SD, AND ND 2002–2009[a]

	Lung					Colorectal					Female Breast					Prostate				
	AI/AN		NHW			AI/AN		NHW			AI/AN		NHW			AI/AN		NHW		
	No.	%	No.	%	X^2/t-test	No.	%	No.	%	X^2/t-test	No.	%	No.	%	X^2/t-test	No.	%	No.	%	X^2/t-test
Age																				
<55	42	10.5	1421	8.9	<.0001	78	28.4	1877	12.8	<.0001	176	43.0	6145	30.7	<.0001	24	8.5	1622	8.6	<.001
55–64	99	24.7	3136	19.7		70	25.5	2485	17.0		120	29.3	4450	22.2		94	33.3	4954	26.2	
65–74	158	39.4	5234	32.8		74	26.9	3604	24.6		62	15.2	4275	21.3		107	37.9	6818	36.1	
75–84	80	20.0	4811	30.2		40	14.5	4366	29.8		39	9.5	3563	17.8		50	17.7	4464	23.6	
85+	22	5.5	1351	8.5		13	4.7	2298	15.7		12	2.9	1606	8.0		7	2.5	1053	5.6	
Mean (SD)	67.7 (10.4)		70.3 (11.0)		<.0001	62.5 (13.4)		71.0 (13.3)		<.0001	57.5 (13.1)		63.5 (14.5)		<.0001	66.7 (9.1)		68.6 (10.0)		<.001
Sex																				
Male	192	47.9	9065	56.8	<.001	143	52.0	7623	52.1	0.97										
Female	209	52.1	6888	43.2		132	48.0	7007	47.9											
SES																				
Very low	208	51.9	261	1.6	<.0001	152	55.3	268	1.8	<.0001	180	44.3	270	1.3	<.0001	164	58.4	374	2.0	<.0001
Low	64	16.0	1305	8.2		31	11.3	1297	8.9		59	14.5	1435	7.2		30	10.7	1841	9.7	
Average	24	6.0	2866	18.0		15	5.5	3017	20.6		34	8.4	3495	17.5		15	5.3	3833	20.3	
High	59	14.7	4691	29.4		48	17.5	4513	30.9		82	20.2	5733	28.6		48	17.1	5709	30.2	
Very high	46	11.5	6828	42.8		29	10.5	5523	37.8		51	12.6	9095	45.4		24	8.5	7135	37.8	
Rurality																				
Urban	76	19.0	7439	46.6	<.0001	50	18.2	5905	40.4	<.0001	93	22.9	9786	48.9	<.0001	48	17.1	7780	41.2	<.0001
Small urban	30	7.5	4111	25.8		28	10.2	4090	28.0		42	10.3	4987	24.9		21	7.5	5104	27.0	
Rural	34	8.5	1402	8.8		26	9.5	1499	10.3		23	5.7	1731	8.6		18	6.4	1783	9.4	
Remote rural	261	65.1	2999	18.8		171	62.2	3124	21.4		248	61.1	3524	17.6		194	69.0	4225	22.4	

Note:
[a]SES and Rurality totals will not equal the total number of cases, due to missing county information (N=48 : Lung=2; CRC=12; BR=14; PR=20).
AI/AN = American Indian/Alaska Native
NHW = Non-Hispanic White
NE = Nebraska
SD = South Dakota
ND = North Dakota
SES = Socioeconomic Status

Table 2.

AGE-ADJUSTED INCIDENCE RATES AND RATE RATIOS FOR OF LUNG, COLORECTAL, FEMALE BREAST, AND PROSTATE CANCER AMONG AI/ANS AND NHWS: NE, SD, AND ND 2002–2009

	Lung					Colorectal					Female Breast					Prostate				
	AI/AN		NHW			AI/AN		NHW			AI/AN		NHW			AI/AN		NHW		
	No.	Rate	No.	Rate	RR	No.	Rate	No.	Rate	RR	No.	Rate	No.	Rate	RR	No.	Rate	No.	Rate	RR
All	401	92.2	15953	60.6	1.5[a]	275	57.0	14630	54.4	1.0	409	134.6	20039	149.3	0.9[a]	282	146.3	18911	158.5	0.9
NE	36	58.4	8861	63.2	0.9	33	41.3	7891	55.2	0.7	36	75.3	11070	152.7	0.5[a]	20	66.5	9565	151.6	0.4[a]
SD	214	89.4	3910	58.1	1.5[a]	159	59.0	3474	50.9	1.2	225	135.4	4812	143.0	0.9	176	160.9	5029	164.0	1.0
ND	151	116.1	3182	57.2	2.0[a]	83	61.2	3265	56.7	1.1	148	166.5	4157	148.0	1.1	86	159.4	4317	169.4	0.9

Note:
[a]RR is significant at .05.
AI/AN = American Indian/Alaska Native
NHW = Non-Hispanic White
NE = Nebraska
SD = South Dakota
ND = North Dakota

considerably among the three states, with Nebraska AI/ANs having consistently lower incidence rates than South or North Dakota AI/ANs. In fact, AI/ANs in Nebraska were at much lower risk of female breast or prostate cancer than NHWs in the state (RR = 0.5 and RR = 0.4, respectively). The combined data from the three states indicate similar age-adjusted incidence rates between AI/ANs and NHWs, with the exception of lung cancer, for which the risk was 1.5 times higher for AI/ANs than for NHWs. In North Dakota, AI/ANs were two times more likely than NHWs to develop lung cancer.

Overall, the AI/AN and NHW staging distributions were similar (Table 3). However, there were significant differences in the age-adjusted incidence rate of late and unstaged cancers (Table 4). AI/ANs were 1.2 times more likely than NHWs to be diagnosed with late-stage cancer and 2.0 times more likely to receive an unstaged cancer diagnosis. Tables 5 and 6 show the results of multivariate analysis for late and unstaged diagnoses. Older age was associated with decreased likelihood of receiving a late-stage diagnosis, and increased likelihood of receiving an unstaged diagnosis. American Indians/Alaska Natives were significantly more likely than NHWs to receive an unstaged diagnosis of cancer but not a late-stage diagnosis. Sex (being male) was a significant predictor for a late-stage diagnosis of lung cancer. Community-level SES was significantly associated with unstaged diagnosis but not with late-stage diagnosis. The likelihood of late-stage diagnosis was higher in more recent years for lung, female breast, and prostate cancers. On the other hand, the likelihood of unstaged diagnosis was lower in more recent years for each of the four cancers studied.

Discussion

Previous publications compiled the evidence for large geographic variations in cancer incidence and mortality within the AI/AN population.[1,25–28] The present study went a step further to examine the data from a three-state region of the Northern Plains with one of the highest incidence and mortality rates among AI/ANs, to investigate the trends and patterns of late and unstaged diagnosis of the four most common types of cancers. Previous analyses of data on the Indian Health Service (IHS) Contract Health Services Delivery Area (CHSDA) counties reported that AI/AN patients tended to be diagnosed with cancer at younger ages.[29–32] The present study examined data from both CHSDA and non-CHSDA counties and found similar results. The difference in age distribution between AI/ANs and NHWs in our study was especially large for colorectal cancer, for which AI/ANs were found to be 8.5 years younger than NHWs (average age: 62.5 *vs.* 71.0 years). In addition, 43% of AI/AN women diagnosed with female breast cancer were younger than 55 years, compared with 30.7% of NHWs. Clinical providers and public health professionals who have roles in educating community members and advising patients should be made aware that AI/ANs are at higher risk of developing cancer at a younger age. Furthermore, more efforts should be made to increase awareness of cancer among younger AI/ANs so that they understand the importance of, and can take the initiative in, leading a healthier lifestyle to decrease the risk of cancer, and following the early detection guidelines.

Other striking differences between AI/AN and NHW cancer patients were observed

TABLE 3.

STAGING DISTRIBUTION FOR AI/ANS AND NHWS WITH LUNG, COLORECTAL, FEMALE BREAST, AND PROSTATE CANCERS: NE, SD, AND ND 2002–2009

	Lung				Colorectal				Female Breast				Prostate			
	AI/AN		NHW		AI/AN		NHW		AI/AN		NHW		AI/AN		NHW	
	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
In situ	0	0.0	10	0.1	12	4.4	656	4.5	52	12.7	3352	16.7	0	0.0	4	0.0
Local	69	17.2	2879	18.0	79	28.7	5389	36.8	193	47.2	10375	51.8	211	74.8	14966	79.1
Regional	89	22.2	3798	23.8	106	38.5	5244	35.8	126	30.8	4998	24.9	25	8.9	2191	11.6
Distant	197	49.1	7566	47.4	53	19.3	2334	16.0	20	4.9	698	3.5	15	5.3	722	3.8
Unstaged	46	11.5	1700	10.7	25	9.1	1007	6.9	18	4.4	616	3.1	31	11.0	1028	5.4

AI/AN = American Indian/Alaska Native
NHW = Non-Hispanic White
NE = Nebraska
SD = South Dakota
ND = North Dakota

Table 4.

AGE-ADJUSTED INCIDENCE RATES AND RATE RATIOS OF LATE AND UNSTAGED LUNG, COLORECTAL, FEMALE BREAST, AND PROSTATE CANCER AMONG AI/ANS AND NHWS: NE, SD, AND ND 2002–2009

	AI/AN		NHWs		Rate Ratio (AI/AN:NHW)	
	Rate	**95% CI**	**Rate**	**95% CI**	**Rate**	**95% CI**
Late stage						
Lung	63.2	55.9–70.5	43.4	42.6–44.2	1.5[a]	1.3–1.6
Colorectal	31.9	26.9–36.9	28.3	27.7–28.9	1.1	0.9–1.3
Female Breast	46.7	39.1–54.3	43.4	42.3–44.5	1.1	0.9–1.3
Prostate	18.9	13.0–24.8	24.0	23.1–24.9	0.8	0.5–1.1
Combined	129.0	118.9–139.1	105.6	104.4–106.8	1.2[a]	1.1–1.3
Unstaged (all cancers)	30.5	25.0–36.0	15.4	14.9–15.9	2.0[a]	1.6–2.4

Note:
[a]RR is significant at .05.
AI/AN = American Indian/Alaska Native
NHW = Non-Hispanic White
NE = Nebraska
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ND = North Dakota
CI = Confidence Interval

for residence and associated SES. For example, 58% of AI/ANs diagnosed with prostate cancer were living in communities classified as having very low SES, compared with 2.0% among NHWs, and 69.0% of AI/ANs were living in remote rural communities at the time of diagnosis with prostate cancer. A similar pattern was observed for patients with other cancer types. Results of a 2012 survey conducted by the Northern Plains Tribal Cancer Data Improvement Initiative showed the limited capacity of tribal and IHS health programs to provide comprehensive cancer prevention and control activities (unpublished data). Access to colonoscopy for colorectal cancer on reservation communities is especially poor in this region. In most cases, diagnostic tests are done through an outside health care system, which delays cancer detection. Many reservation communities are located hundreds of miles away from the nearest hospital that provides cancer screening and diagnostic services, and if the tribe does not have funds, the patient may not be covered for the cost of tests.

A surprising finding of this study was the large variation in incidence rates across the three states examined. Regardless of cancer type, within the AI/AN population, in most cases North Dakota had the highest age-adjusted incidence rate, followed by South Dakota. Nebraska AI/ANs not only had much lower rates than North or South Dakota AIs, but they also had rates lower than Nebraska NHWs for lung, colorectal,

Table 5.

MULTIVARIATE REGRESSION FOR LATE STAGE CANCERS FOR AI AND NHWS WITH LUNG, COLORECTAL, FEMALE BREAST, AND PROSTATE CANCERS: NE, SD, AND ND 2002–2009

	Lung			Colorectal			Female Breast			Prostate		
	OR	95% CI	p-value	OR	95% CI	p-value	OR	95% CI	p-value	OR	95% CI	p-value
Age												
<55	1.0			1.0			1.0			1.0		
55–64	0.8	0.7–1.0	.04	1.0	0.9–1.1	.97	0.8	0.7–0.9	<.0001	1.0	0.8–1.1	.64
65–74	0.6	0.5–0.7	<.0001	0.8	0.7–0.9	<.01	0.7	0.7–0.8	<.0001	0.7	0.7–0.9	<.0001
75–84	0.6	0.5–0.7	<.0001	0.8	0.7–0.9	<.01	0.7	0.6–0.8	<.0001	0.5	0.4–0.6	<.0001
85+	0.6	0.5–0.7	<.0001	0.9	0.8–1.0	.18	0.9	0.8–1.0	.08	1.1	0.9–1.4	.36
		<0.0001			0.0041			<0.0001			<0.0001	
Race												
NHW	1.0			1.0			1.0			1.0		
AI	0.9	0.7–1.2	.53	1.3	1.0–1.7	.08	1.3	1.0–1.7	.02	1.1	0.8–1.7	.50
Sex												
Male	1.0			1.0								
Female	0.8	0.7–0.9	<.0001	1.0	1.0–1.1	.64						
SES												
Very high	1.0			1.0			1.0			1.0		
High	0.9	0.8–1.1	.42	1.0	0.9–1.1	.98	1.0	0.9–1.1	.83	0.9	0.8–1.0	.12
Average	0.9	0.7–1.1	.18	1.0	0.9–1.2	.90	1.0	0.9–1.2	.53	0.9	0.7–1.0	.10
Low	1.1	0.8–1.4	.61	1.0	0.8–1.2	.80	1.1	0.8–1.2	.83	0.9	0.7–1.1	.23
Very low	1.2	0.8–1.7	.37	1.1	0.8–1.4	.65	1.0	0.8–1.4	.61	0.6	0.4–0.8	<.01
		0.0851			0.7266			0.0017			0.0027	

(*Continued on p. 1058*)

Table 5. (*continued*)

	Lung			Colorectal			Female Breast			Prostate		
	OR	95% CI	p-value	OR	95% CI	p-value	OR	95% CI	p-value	OR	95% CI	p-value
Rurality												
Urban	1.0			1.0			1.0			1.0		
Small urban	1.1	0.9–1.3	.27	0.9	0.7–1.0	.02	1.1	1.0–1.3	.04	0.9	0.8–1.1	.28
Rural	1.1	0.9–1.4	.24	1.1	0.9–1.3	.19	1.1	1.0–1.3	.18	1.3	1.0–1.5	.03
Remote rural	1.1	0.9–1.4	.25	1.0	0.8–1.1	.66	1.1	1.0–1.3	.12	1.1	0.9–1.3	.28
		0.1185			0.7046			0.0027			0.2049	
Years of diagnosis												
2002–2005	1.0			1.0			1.0			1.0		
2006–2009	1.2	1.1–1.3	<.001	0.9	0.9–1.0	.06	1.0	1.0–1.1	.16	1.1	1.0–1.2	<.01

AI = American Indian
NHW = Non-Hispanic White
NE = Nebraska
SD = South Dakota
ND = North Dakota
OR = Odds Ratio
CI = Confidence Interval

Table 6.

MULTIVARIATE REGRESSION FOR UNSTAGED CANCERS FOR AI AND NHWS WITH LUNG, COLORECTAL, FEMALE BREAST, AND PROSTATE CANCERS: NE, SD, AND ND 2002–2009

	Lung			Colorectal			Female Breast			Prostate		
	OR	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**
Age												
<55	1.0			1.0			1.0			1.0		
55–64	1.1	0.8–1.4	.56	0.8	0.6–1.0	.07	1.1	0.8–1.5	.47	0.8	0.6–1.2	.27
65–74	1.3	1.0–1.7	.03	1.0	0.8–1.3	.91	1.2	0.9–1.6	.23	1.1	0.8–1.6	.52
75–84	2.3	1.8–2.9	<.0001	1.3	1.0–1.7	.02	2.0	1.6–2.7	<.0001	3.3	2.4–4.6	<.0001
85+	6.4	5.0–8.2	<.0001	4.1	3.2–5.2	<.0001	8.7	6.8–11.0	<.0001	14.1	10.0–19.8	<.0001
		<0.0001			<0.0001			<0.0001			<0.0001	
Race												
NHW	1.0			1.0			1.0			1.0		
AI	1.2	0.9–1.8	.23	1.6	1.0–2.7	.04	1.8	1.0–3.1	.04	2.3	1.4–3.6	<.001
Sex												
Male	1.0			1.0								
Female	0.9	0.9–1.1	.31	1.0	0.9–1.1	1.00						
SES												
Very high	1.0			1.0			1.0			1.0		
High	1.4	1.2–1.7	<.001	1.2	0.9–1.6	.19	1.2	0.9–1.6	.24	1.7	1.4–2.2	<.0001
Average	1.6	1.3–2.1	<.0001	1.4	1.0–1.9	.04	2.1	1.5–3.1	<.0001	1.9	1.4–2.6	<.0001
Low	1.3	1.0–1.7	.08	1.4	1.0–2.0	.08	1.8	1.1–2.8	.01	1.7	1.2–2.4	<.01
Very low	1.2	0.8–1.9	.33	1.4	0.8–2.2	.22	1.9	1.0–3.5	.06	2.3	1.4–3.6	<.001
		<0.0001			<0.0001			<0.0001			<0.0001	

(*Continued on p. 1060*)

Table 6. (*continued*)

	Lung			Colorectal			Female Breast			Prostate		
	OR	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**
Rurality												
Urban	1.0			1.0			1.0			1.0		
Small urban	0.8	0.7–1.0	.03	0.8	0.6–1.0	.05	0.6	0.4–0.9	<.01	0.5	0.4–0.6	<.0001
Rural	1.1	0.9–1.5	.28	1.0	0.7–1.4	.90	0.7	0.5–1.1	.13	0.6	0.4–0.8	<.01
Remote rural	0.9	0.7–1.2	.65	1.0	0.7–1.3	.77	0.7	0.5–1.0	.04	0.6	0.5–0.9	<.01
		<0.0001			<0.0001			<0.0001			<0.0001	
Years of diagnosis												
2002–2005	1.0			1.0			1.0			1.0		
2006–2009	0.6	0.5–0.6	<.0001	0.7	0.6–0.8	<.0001	0.5	0.4–0.5	<.0001	0.4	0.4–0.5	<.0001

AI = American Indian
NHW = Non-Hispanic White
NE = Nebraska
SD = South Dakota
ND = North Dakota
OR = Odds Ratio
CI = Confidence Interval

and prostate cancers. For instance, the age-adjusted incidence rate of female breast and prostate cancer among Nebraska AI/ANs was about half of that of Nebraska NHWs. Due to the relatively small number of cases in Nebraska, the incidence rate estimates are less reliable than for North Dakota or South Dakota. Nevertheless, it is clear that there is a distinct and consistent pattern of cancer incidence rates for Nebraska.

There are several possible reasons for the observed variations in cancer incidence rates between Nebraska and other two states. One possibility is that AI/ANs in Nebraska have less exposure to cancer risk factors. For example, Behavioral Risk Factor Surveillance System data indicate that AI/ANs in Nebraska have a much lower smoking rate and are less obese than AI/ANs in North and South Dakota.[33–35] A higher cancer screening rate among AI/ANs in Nebraska than in North and South Dakota may also contribute to variation in incidence rates.[33–35] There could also be a difference in physical environment that results in differences in exposure to environmental carcinogens. Another possible cause of variation in incidence rates is under-reporting of cancer among AI/ANs in Nebraska. A connection between IHS and cancer registry data is done routinely in all three states. However, the connection is not effective if AI/AN patients do not use IHS health care and are therefore not in the IHS database. There are a variety of reasons why AI/AN individuals do not use IHS health care. Some AI/AN individuals with private insurance may choose to use a non-IHS health care system, and other AI/AN individuals may not be eligible to receive IHS health care. Future research should look into reasons for incidence rate variations across states so that specific recommendations can be made to improve the data quality and/or to decrease the exposure to cancer risk factors.

As has been found in previously published research,[29–32] the present study found that AI/ANs are more likely than NHWs to receive a late-stage diagnosis or an unstaged diagnosis. We conducted a multivariate analysis to identify factors that are independently associated with late- or early-stage diagnosis. In this study, being older was found to be a risk factor for late diagnosis of all four cancers. Previous research offers mixed results on the effect of age, and the results differ by type of cancer. For example, older age was associated with delayed presentation with symptoms of female breast cancer[36] but was unrelated to delayed presentation for colorectal cancer[37] and lung cancer.[38,39] After adjusting for other variables, in this study, race was no longer a predictor for late-stage diagnosis of cancer. However, other studies found non-White ethnic origin to be a risk factor for longer delay in presenting with female breast[36] or prostate cancer.[40] This study found that males were 20% more likely than females to receive a late-stage diagnosis for lung cancer, but other studies did not find such an association.[38,39,41,42] Previous studies found that lower SES was associated with late-stage diagnosis among men with prostate cancer,[40,43,44] but no overall relationship between SES and late-stage diagnosis has been found for colorectal cancer[37] or lung cancer.[38,39,43] The present study found no association between SES and late-stage diagnosis.

Multivariate analysis results indicate that the reasons for receiving an unstaged or a late-stage diagnosis differ. Older age was associated with unstaged diagnosis but not with late-stage diagnosis. This result was confirmed by previous research.[45–49] Additionally, lower community SES was associated with likelihood of unstaged diagnosis but not with late-stage diagnosis. Again, this result was replicated in previous research.[46,48–50] Another difference between the results for late-stage and unstaged diagnoses was that AI/ANs were significantly more likely to receive an unstaged diagnosis even after adjusting for other factors. Previous studies found an association between being a racial/ethnic minority and receiving an unstaged cancer diagnosis.[11,48,51]

The following strengths and limitations of this study should be considered. The cancer registries of Nebraska and North and South Dakota have all met the data quality standard of the North American Association of Central Cancer Registries in terms of completeness of capturing cancer cases and accuracy of information. These registries also have data-sharing agreements in order to obtain information about individuals who receive cancer care outside of their state of residence. Because it was outside of the scope of study, we did not look into the extent of racial misclassification in these cancer registries. Therefore, it was not possible to determine if or how racial misclassification may have influenced our results regarding the variations in cancer incidence rates across the states. Though cancer registries collect high-quality data on cancer staging and diagnosis, they do not consistently collect demographic information beyond age and sex. Therefore, we used U.S. Census Bureau data to determine the SES of cancer patients. Self-reported individualized information on education, marital status, and income may give a more accurate picture, but such information was not available.

In summary, compared with NHWs, AI/ANs in a three-state area of the Northern Plains are diagnosed with cancer at a much earlier age and are more likely to live in rural communities classified as having low SES. An important finding of the study was wide variation in cancer incidence rates across the three states, a finding that warrants future research on cancer surveillance and epidemiology research. Overall, AI/ANs were 20% more likely than NHWs to receive a late-stage diagnosis and two times more likely to receive an unstaged diagnosis. In order to understand the reasons for late-stage and unstaged diagnosis of cancer, future research should examine more detailed information on patients and the health care system. The effect of late and unstaged diagnosis of cancer on survival should also be investigated.

References

1. Espey DK, Wiggins CL, Jim MA, et al. Methods for improving cancer surveillance data in American Indian and Alaska Native populations. Cancer. 2008 Sep 1;113(5 Suppl):1120–30.
http://dx.doi.org/10.1002/cncr.23724
PMid:18720372
2. Espey D, Paisano R, Cobb N. Regional patterns and trends in cancer mortality among American Indians and Alaska Natives, 1990–2001. Cancer. 2005 Mar 1;103(5):1045–53.
http://dx.doi.org/10.1002/cncr.20876
PMid:15685622
3. Norris T, Vines PL, Hoeffel EM. The American Indian and Alaska Native population: 2010. U.S. Census Bureau. 2012 Jan;1–21.
4. Liu L, Deapen D, Bernstein L. Socioeconomic status and cancers of the female breast and reproductive organs: a comparison across racial/ethnic populations in Los Angeles County, California (United States). Cancer Causes Control. 1998 Aug 1;9(4):369–80.
http://dx.doi.org/10.1023/A:1008811432436
PMid:9794168

5. Yost K, Perkins C, Cohen R, et al. Socioeconomic status and breast cancer incidence in California for different race/ethnic groups. Cancer Causes Control. 2001 Oct;12(8):703–11.
http://dx.doi.org/10.1023/A:1011240019516
PMid:11562110
6. Blair SL, Sadler GR, Bristol R, et al. Early cancer detection among rural and urban Californians. BMC Public Health. 2006 Jul 26;6(194):1–5.
7. Campbell NC, Elliott AM, Sharp L, et al. Rural and urban differences in stage at diagnosis of colorectal and lung cancers. Br J Cancer. 2001 Apr 6;84(7):910–14.
http://dx.doi.org/10.1054/bjoc.2000.1708
PMid:11286470 PMCid:PMC2363829
8. Hawley ST, Chang S, Risser D, et al. Colorectal cancer incidence and mortality in Texas 1990–1992: a comparison of rural classifications. J Rural Health. 2002 Fall;18(4):536–46.
http://dx.doi.org/10.1111/j.1748-0361.2002.tb00920.x
PMid:12380896
9. Higginbotham JC, Moulder J, Currier M. Rural v. urban aspects of cancer: first-year data from the Mississippi Central Cancer Registry. Fam Community Health. 2001 Jul;24(2):1–9.
http://dx.doi.org/10.1097/00003727-200107000-00003
PMid:11373161
10. Sankaranarayanan J, Watanabe-Galloway S, Sun J, et al. Rurality and other determinants of early colorectal cancer diagnosis in Nebraska: a 6-year cancer registry study, 1998–2003. J Rural Health. 2009 Fall;25(4):358–65.
http://dx.doi.org/10.1111/j.1748-0361.2009.00244.x
PMid:19780915
11. Merrill RM, Sloan A, Anderson AE, et al. Unstaged cancer in the United States: a population-based study. BMC Cancer. 2011 Sep 21;11(402):1–10.
http://dx.doi.org/10.1186/1471-2407-11-402
12. Cetin K, Beebe-Dimmer JL, Fryzek JP, et al. Recent time trends in the epidemiology of stage IV prostate cancer in the United States: analysis of data from the Surveillance, Epidemiology, and End Results program. Urology. 2010 Jun;75(6):1396–404.
http://dx.doi.org/10.1016/j.urology.2009.07.1360
PMid:19969335 PMCid:PMC4249683
13. Li J, Djenaba JA, Soman A, et al. Recent trends in prostate cancer incidence by age, cancer stage, and grade, the United States, 2001–2007. Prostate Cancer. 2012;2012(691380):1–8.
http://dx.doi.org/10.1155/2012/691380
PMid:23251806 PMCid:PMC3515924
14. Akinyemiju TF, Soliman AS, Copeland G, et al. Trends in breast cancer stage and mortality in Michigan (1992–2009) by race, socioeconomic status, and area healthcare resources. PLoS One. 2013 Apr 29;8(4):e61879.
http://dx.doi.org/10.1371/journal.pone.0061879
PMid:23637921 PMCid:PMC3639257
15. American Cancer Society. Cancer facts & figures 2013. Atlanta, GA: American Cancer Society, 2013. Available at: http://www.cancer.org/acs/groups/content/@epidemiology surveilance/documents/document/acspc-036845.pdf.
16. Ghelfi L, Parker T. Developing a county-level measure of urban influence. Amber Waves. 2004 Apr;2(2):1–4.

17. Krieger N. Overcoming the absence of socioeconomic data in medical records: validation and application of a census-based methodology. Am J Public Health. 1992 May;82(5):703–10.
http://dx.doi.org/10.2105/AJPH.82.5.703
PMid:1566949 PMCid:PMC1694121
18. Roetzheim RG, Pal N, Tennant C, et al. Effects of health insurance and race on early detection of cancer. J Natl Cancer Inst. 1999 Aug 18;91(16):1409–15.
http://dx.doi.org/10.1093/jnci/91.16.1409
PMid:10451447
19. Sankaranarayanan J, Watanabe-Galloway S, Sun J, et al. Age and rural residence effects on accessing colorectal cancer treatments: a registry study. Am J Manag Care. 2010 Apr;16(4):265–73.
PMid:20394462
20. Hall SA, Kaufman JS, Ricketts TC. Defining urban and rural areas in U.S. epidemiologic studies. J Urban Health. 2006 Mar;83(2):162–75.
http://dx.doi.org/10.1007/s11524-005-9016-3
PMid:16736366 PMCid:PMC2527174
21. Parker T, Ghelfi LM. Using the 2003 Urban Influence Codes to understand rural America. Washington, DC: United States Department of Agriculture, 2004. Available at: http://www.ers.usda.gov/amber-waves/2004-april/using-the-2003-urban-influence-codes-to-understand-rural-america.aspx#.VWS4As9VhBc.
22. Cramer ME, Chen LW, Mueller KJ, et al. Predictive model to determine need for nursing workforce. Policy, Politics Nurs Pract. 2004 Aug;5(3):174–90.
http://dx.doi.org/10.1177/1527154404266785
23. Klein RJ, Schoenborn CA. Age adjustment using the 2000 projected U.S. population. Healthy People 2010. 2001 Jan;20:1–10.
24. National Cancer Institute. Surveillance, Epidemiology, and End Results (SEER) Program. U.S. population data 1969–2013. Bethesda, MD: National Cancer Institute, 2014. Available at: http://seer.cancer.gov/resources.
25. Haverkamp D, Espey D, Paisano R, et al. Cancer mortality among American Indians and Alaska Natives: regional differences, 1999–2003. Rockville, MD: Indian Health Service, 2008. Available at: http://www.ihs.gov/epi/documents/cancer/9903CMAIANRD.pdf.
26. Cobb N, Paisano RE. Patterns of cancer mortality among Native Americans. Cancer. 1998 Dec 1;83(11):2377–83.
http://dx.doi.org/10.1002/(SICI)1097-0142(19981201)83:11<2377::AID-CNCR18>3.0.CO;2-Z
27. Centers for Disease Control and Prevention (CDC). Cancer mortality among American Indians and Alaska Natives—United States, 1994–1998. MMWR Morb Mortal Wkly Rep. 2003 Aug 1;52(30):704–07.
PMid:12894057
28. Nutting PA, Freeman WL, Risser DR, et al. Cancer incidence among American Indians and Alaska Natives, 1980 through 1987. Am J Public Health. 1993 Nov;83(11):1589–98.
http://dx.doi.org/10.2105/AJPH.83.11.1589
PMid:8238684 PMCid:PMC1694894
29. Bliss A, Cobb N, Solomon T, et al. Lung cancer incidence among American Indians and Alaska Natives in the United States, 1999–2004. Cancer. 2008 Sep 1;113(5 Suppl):1168–78.

http://dx.doi.org/10.1002/cncr.23738
PMid:18720387
30. Wingo PA, King J, Swan J, et al. Breast cancer incidence among American Indian and Alaska Native women: U.S., 1999–2004. 2008 Sep 1;113(5 Suppl):1191–202.
31. Henderson JA, Espey DK, Jim MA, et al. Prostate cancer incidence among American Indian and Alaska Native men, U.S., 1999–2004. Cancer. 2008 Sep 1;113(5 Suppl):1203–12.
http://dx.doi.org/10.1002/cncr.23739
PMid:18720376
32. Perdue DG, Perkins C, Jackson-Thompson J, et al. Regional differences in colorectal cancer incidence, stage, and subsite among American Indians and Alaska Natives, 1999–2004. Cancer. 2008 Sep 1;113(5 Suppl):1179–90.
http://dx.doi.org/10.1002/cncr.23726
PMid:18720388
33. Great Plains Tribal Chairmen's Health Board. Cancer health status report for Nebraska American Indian communities, 2012. Rapid City, SD: Great Plains Tribal Chairmen's Health Board, 2012.
34. Great Plains Tribal Chairmen's Health Board and North Dakota Department of Health. Cancer health status report for North Dakota American Indian communities, 2012. Rapid City, SD: Great Plains Tribal Chairmen's Health Board, 2012.
35. Great Plains Tribal Chairmen's Health Board and South Dakota Department of Health. Cancer health status report for South Dakota American Indian communities, 2011. Rapid City, SD: Great Plains Tribal Chairmen's Health Board, 2011. Available at: http://getscreened.sd.gov/documents/AI-CancerHealthStatus.pdf.
36. Ramirez AJ, Westcombe AM, Burgess CC, et al. Factors predicting delayed presentation of symptomatic breast cancer: a systematic review. Lancet. 1999 Apr 3;353(9159):1127–31.
http://dx.doi.org/10.1016/S0140-6736(99)02142-X
37. Mitchell E, Macdonald S, Campbell NC, et al. Influences on pre-hospital delay in the diagnosis of colorectal cancer: a systematic review. Br J Cancer. 2008 Jan 15;98(1):60–70.
http://dx.doi.org/10.1038/sj.bjc.6604096
PMid:18059401 PMCid:PMC2359711
38. Worden JW, Weisman AD. Psychosocial components of lagtime in cancer diagnosis. J Psychosom Res. 1975 Feb;19(1):69–79.
http://dx.doi.org/10.1016/0022-3999(75)90052-5
39. Mor V, Masterson-Allen S, Goldberg R, et al. Pre-diagnostic symptom recognition and help seeking among cancer patients. J Community Health. 1990 Aug;15(4):253–66.
http://dx.doi.org/10.1007/BF01350291
PMid:2212095
40. Samet JM, Hunt WC, Lerchen ML, et al. Delay in seeking care for cancer symptoms: a population-based study of elderly New Mexicans. J Natl Cancer Inst. 1988 May 18;80(6):432–38.
http://dx.doi.org/10.1093/jnci/80.6.432
PMid:3367383
41. Risberg T, Sørbye, SW, Norum J, et al. Diagnostic delay causes more psychological distress in female than in male cancer patients. Anticancer Res. 1996 Mar–Apr;16(2):995–99.
PMid:8687166

42. Bowen EF, Rayner CFJ. Patient and GP led delays in the recognition of symptoms suggestive of lung cancer. Lung Cancer. 2002 Aug;37(2):227–28.
http://dx.doi.org/10.1016/S0169-5002(02)00143-5
43. Hackett TP, Cassem NH, Raker JW. Patient delay in cancer. N Engl J Med. 1973 Jul 5;289(1):14–20.
http://dx.doi.org/10.1056/NEJM197307052890104
PMid:4708950
44. Fitzpatrick P, Corcoran N, Fitzpatrick JM. Prostate cancer: How aware is the public? Br J Urol. 1998 Jul;82(1):43–48.
http://dx.doi.org/10.1046/j.1464-410x.1998.00685.x
PMid:9698661
45. Yancik R, Wesley MN, Ries LA, et al. Effect of age and comorbidity in postmenopausal breast cancer patients aged 55 years and older. JAMA. 2001 Feb 21;285(7):885–92.
http://dx.doi.org/10.1001/jama.285.7.885
PMid:11180731
46. Klassen AC, Curriero F, Kulldorff M, et al. Missing stage and grade in Maryland prostate cancer surveillance data, 1992–1997. Am J Prev Med. 2006 Feb;30(2 Suppl):S77–87.
http://dx.doi.org/10.1016/j.amepre.2005.09.010
PMid:16458794
47. Koroukian SM, Xu F, Beaird H, et al. Complexity of care needs and unstaged cancer in elders: a population-based study. Cancer Detect Prev. 2007;31(3):199–206.
http://dx.doi.org/10.1016/j.cdp.2007.04.002
PMid:17658225 PMCid:PMC2577170
48. Worthington JL, Koroukian SM, Cooper GS. Examining the characteristics of unstaged colon and rectal cancer cases. Cancer Detect Prev. 2008;32(3):251–58.
http://dx.doi.org/10.1016/j.cdp.2008.08.006
PMid:18804920
49. Bradley CJ, Clement JP, Lin C. Absence of cancer diagnosis and treatment in elderly Medicaid-insured nursing home residents. J Natl Cancer Inst. 2008 Jan 2;100(1):21–31.
http://dx.doi.org/10.1093/jnci/djm271
PMid:18159068
50. Roetzheim RG, Pal N, Gonzalez EC, et al. Effects of health insurance and race on colorectal cancer treatments and outcomes. Am J Public Health. 2000 Nov;90(11):1746–54.
http://dx.doi.org/10.2105/AJPH.90.11.1746
PMid:11076244 PMCid:PMC1446414
51. Miller BA, Hankey BF, Thomas TL. Impact of sociodemographic factors, hormone receptor status, and tumor grade on ethnic differences in tumor stage and size for breast cancer in U.S. women. Am J Epidemiol. 2002 Mar 15;155(6):534–45.
http://dx.doi.org/10.1093/aje/155.6.534
PMid:11882527